

Simulation Plant and Excitation Comparison Report

Generated from the existing dashboard model on 2026-06-03 20:36:26

1. What Was Run

The existing simulation model was run for every plant P01-P10 and every excitation profile ET1, ET3, ET6, E_Toggle, and EVR. No model equations, plant values, excitation functions, or dashboard UI files were changed. Each run used the Simulation tab defaults: 4 s duration, 1 ms RK4 physics step, 10 ms controller update, 10 ms logging, no sensor noise, and the plant's current recommended excitation amplitude.

Important model note: the current extracted reference applies plant-specific EA. The per-roller R, J, f, L, and b arrays remain the current baseline arrays unless those exact arrays are supplied in the project reference.

2. Plant Definitions and Differences

Plant	Material	Scale	EA (N)	Regime	ζ CL,min	Paper OS%	Recommended amp (V)	Dynamic?
P01	PE	lab	3,200	O-UD	0.151	45.80	0.08	yes
P02	PET	lab	9,600	O-UD	0.237	45.30	0.04619	yes
P03	PET	lab	40,000	H-Osc	0.35	28.80	0	zero input
P04	PET	pilot	50,000	O-UD	0.201	41.00	0	zero input
P05	Paper	prod	144,000	O-UD	0.144	50.10	0	zero input
P06	Al	lab	310,500	H-Damp	0.515	9.8	0	zero input
P07	Al	pilot	414,000	H-Osc	0.357	31.80	0	zero input
P08	Al	prod	1.242e+06	O-UD	0.179	55.30	0	zero input
P09	Cu	lab	351,000	H-Damp	0.53	16.10	0	zero input
P10	Cu	pilot	351,000	O-UD	0.21	51.90	0	zero input

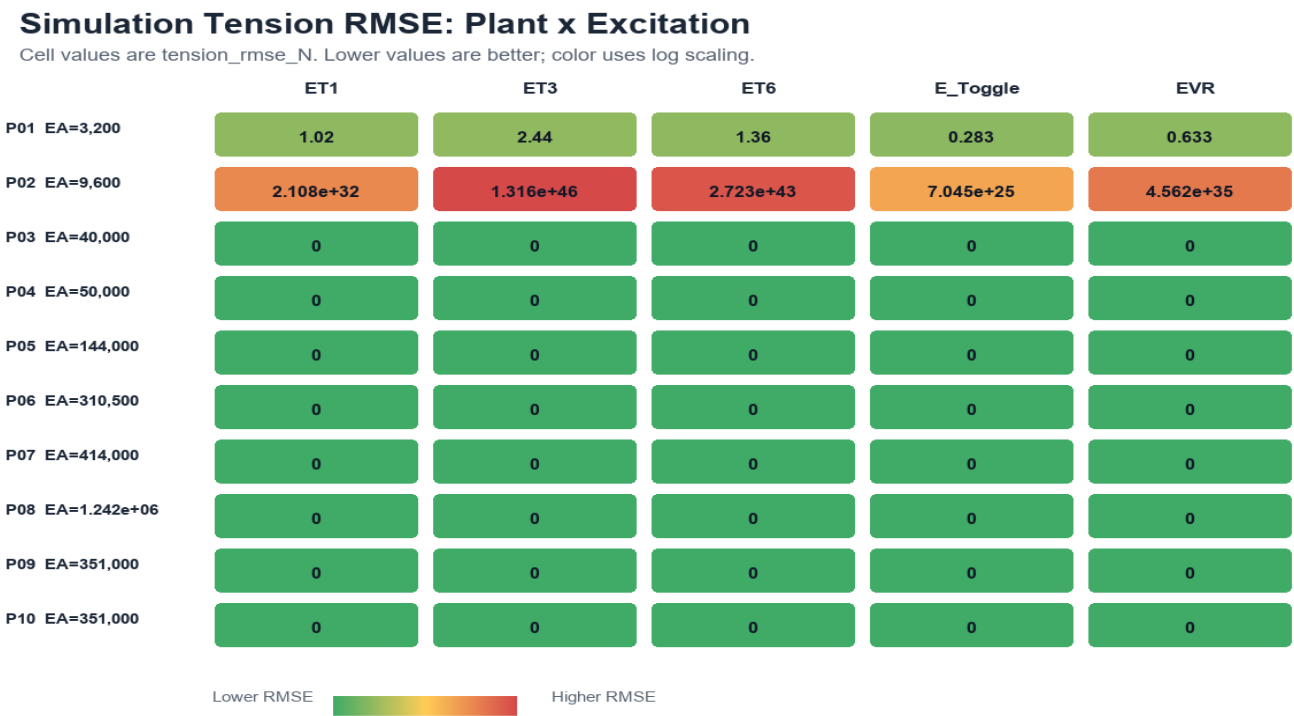
Plants differ by material, production scale, axial stiffness EA, damping/regime label, and supplement overshoot value. In this dashboard model run, EA is the plant parameter that is numerically applied to the simulator.

3. Excitation Definitions and Changing Factors

Excitation	Type	Channels	Changing factors
ET1	single-channel sine	UW only	0.70 Hz sine on u_UW; u_Nip and u_RW are zero.
ET3	three-channel sine	UW, Nip, RW	0.55, 0.80, and 1.10 Hz sine waves with phase offsets.
ET6	three-channel multi-sine	UW, Nip, RW	Two sine components per motor input; richer frequency content than ET3.
E_Toggle	three-channel square/toggle	UW, Nip, RW	Square waves with staggered 0.42, 0.58, and 0.74 s periods.
EVR	event-varying random	UW, Nip, RW	Random held voltage values updated in 0.15 s buckets.

Changing excitation changes waveform type, active motor channels, frequency content, phase/staggering, and how much the web-tension states move during the run. That is why the metrics are different when the excitation type changes.

4. Graph Comparison



Note: P03-P10 use zero recommended excitation in the current dashboard model because EA is outside the extracted baseline range.

Figure 1. Plant x excitation heatmap using tension RMSE.

Best Simulation Result by Plant

Lowest tension_rmse_N across the five excitation choices; log scale keeps zero-input and outlier cases visible.

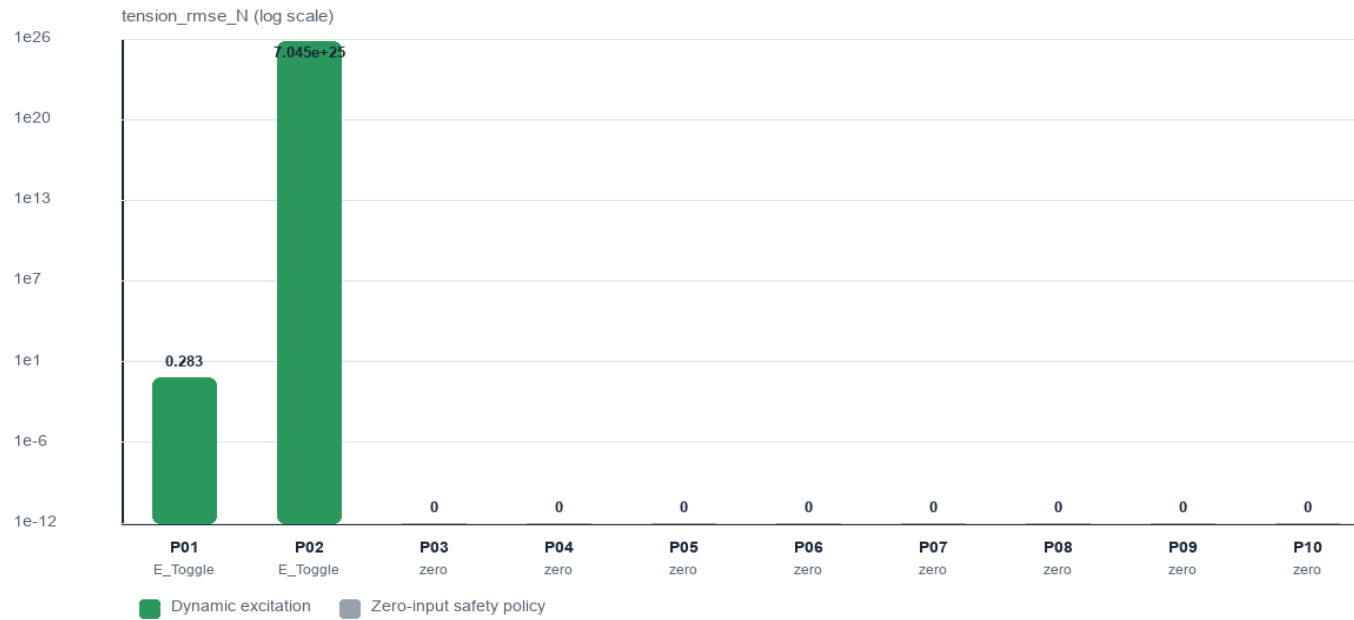


Figure 2. Best excitation choice by plant.

Average Tension RMSE by Excitation

Grouped bars compare all plants to dynamic-amplitude cases only.

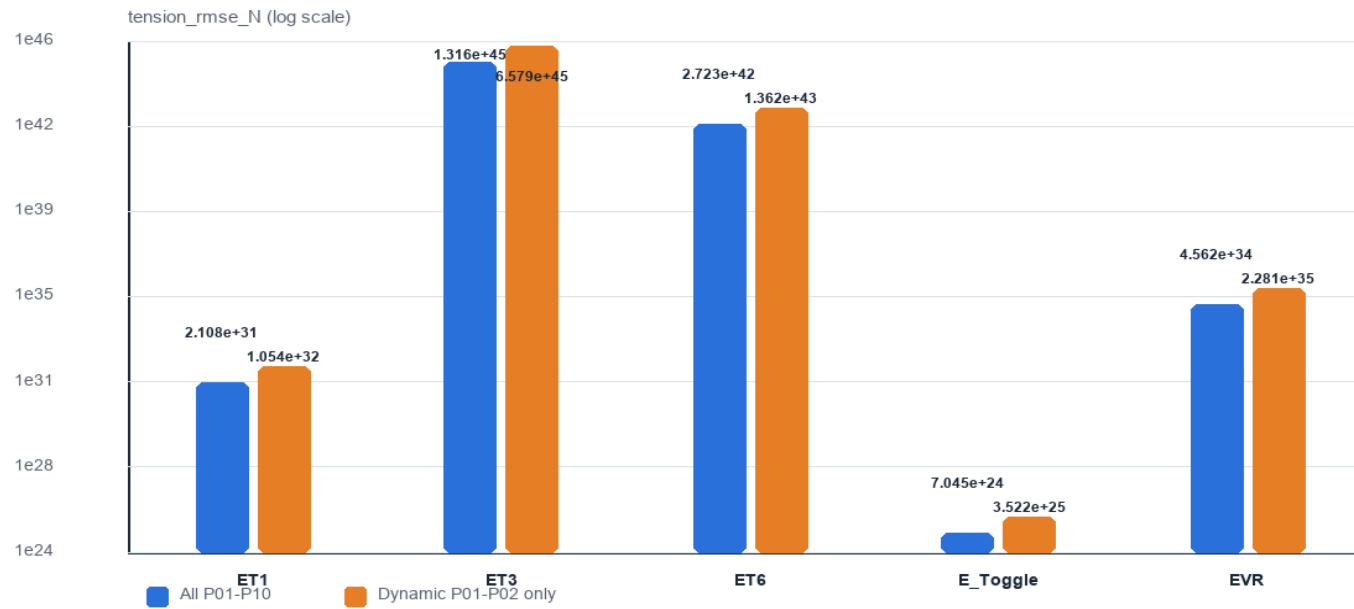


Figure 3. Average tension RMSE by excitation.

Plant EA vs Best Tension RMSE

This shows how plant stiffness maps to the best available simulation error in the current dashboard model.

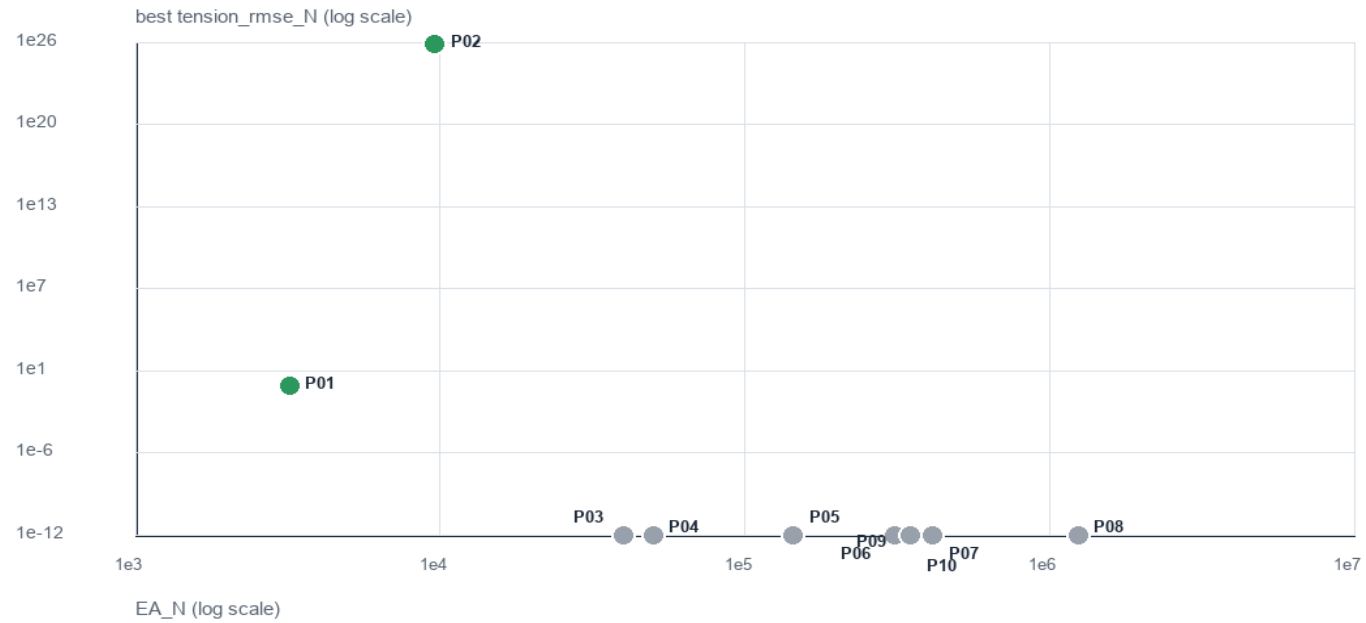


Figure 4. Plant EA compared with best available tension RMSE.

5. Best Result by Plant

Plant	EA (N)	Regime	Amp (V)	Best excitation	Best RMSE (N)	Overshoot (N)	Control RMS (V)
P01	3,200	O-UD	0.08	E_Toggle	0.283	0.7565	2.405
P02	9,600	O-UD	0.04619	E_Toggle	7.045e+25	1.134e+27	15.70
P03	40,000	H-Osc	0	zero-input policy	0	0	2.392
P04	50,000	O-UD	0	zero-input policy	0	0	2.392
P05	144,000	O-UD	0	zero-input policy	0	0	2.392
P06	310,500	H-Damp	0	zero-input policy	0	0	2.392
P07	414,000	H-Osc	0	zero-input policy	0	0	2.392
P08	1.242e+06	O-UD	0	zero-input policy	0	0	2.392
P09	351,000	H-Damp	0	zero-input policy	0	0	2.392
P10	351,000	O-UD	0	zero-input policy	0	0	2.392

6. Full Simulation Table

Plant	Excitation	Amp (V)	Input	Tension RMSE (N)	Max overshoot (N)	t90 (s)	Control RMS (V)
P01	ET1	0.08	dynamic excitation	1.019	2.843	0	2.565
P01	ET3	0.08	dynamic excitation	2.437	6.062	0	3.26
P01	ET6	0.08	dynamic excitation	1.357	3.817	0	2.692
P01	E_Toggle	0.08	dynamic excitation	0.283	0.7565	0	2.405
P01	EVR	0.08	dynamic excitation	0.6326	1.881	0	2.479
P02	ET1	0.04619	dynamic excitation	2.108e+32	3.392e+33	0	17.96
P02	ET3	0.04619	dynamic excitation	1.316e+46	2.117e+47	0	20.39
P02	ET6	0.04619	dynamic excitation	2.723e+43	4.382e+44	0	19.94
P02	E_Toggle	0.04619	dynamic excitation	7.045e+25	1.134e+27	0	15.70
P02	EVR	0.04619	dynamic excitation	4.562e+35	7.341e+36	0	18.31
P03	ET1	0	zero excitation	0	0	0	2.392
P03	ET3	0	zero excitation	0	0	0	2.392
P03	ET6	0	zero excitation	0	0	0	2.392
P03	E_Toggle	0	zero excitation	0	0	0	2.392
P03	EVR	0	zero excitation	0	0	0	2.392
P04	ET1	0	zero excitation	0	0	0	2.392
P04	ET3	0	zero excitation	0	0	0	2.392
P04	ET6	0	zero excitation	0	0	0	2.392
P04	E_Toggle	0	zero excitation	0	0	0	2.392
P04	EVR	0	zero excitation	0	0	0	2.392
P05	ET1	0	zero excitation	0	0	0	2.392

Plant	Excitation	Amp (V)	Input	Tension RMSE (N)	Max overshoot (N)	t90 (s)	Control RMS (V)
P05	ET3	0	zero excitation	0	0	0	2.392
P05	ET6	0	zero excitation	0	0	0	2.392
P05	E_Toggle	0	zero excitation	0	0	0	2.392
P05	EVR	0	zero excitation	0	0	0	2.392
P06	ET1	0	zero excitation	0	0	0	2.392
P06	ET3	0	zero excitation	0	0	0	2.392
P06	ET6	0	zero excitation	0	0	0	2.392
P06	E_Toggle	0	zero excitation	0	0	0	2.392
P06	EVR	0	zero excitation	0	0	0	2.392
P07	ET1	0	zero excitation	0	0	0	2.392
P07	ET3	0	zero excitation	0	0	0	2.392
P07	ET6	0	zero excitation	0	0	0	2.392
P07	E_Toggle	0	zero excitation	0	0	0	2.392
P07	EVR	0	zero excitation	0	0	0	2.392
P08	ET1	0	zero excitation	0	0	0	2.392
P08	ET3	0	zero excitation	0	0	0	2.392
P08	ET6	0	zero excitation	0	0	0	2.392
P08	E_Toggle	0	zero excitation	0	0	0	2.392
P08	EVR	0	zero excitation	0	0	0	2.392
P09	ET1	0	zero excitation	0	0	0	2.392
P09	ET3	0	zero excitation	0	0	0	2.392
P09	ET6	0	zero excitation	0	0	0	2.392
P09	E_Toggle	0	zero excitation	0	0	0	2.392
P09	EVR	0	zero excitation	0	0	0	2.392
P10	ET1	0	zero excitation	0	0	0	2.392
P10	ET3	0	zero excitation	0	0	0	2.392
P10	ET6	0	zero excitation	0	0	0	2.392
P10	E_Toggle	0	zero excitation	0	0	0	2.392
P10	EVR	0	zero excitation	0	0	0	2.392

7. Main Interpretation

Changing plant changes the simulation because plant EA changes the web stiffness and therefore the speed-to-tension coupling. The current project reference also labels plants by material, scale, closed-loop damping regime, and paper overshoot percentage.

Changing excitation changes the simulation because each excitation has a different input waveform, active channel set, frequency content, and phase/staggering. These differences change how strongly each roller and span is disturbed.

P01 and P02 are the only plants inside the extracted baseline EA range, so the dashboard gives them nonzero recommended excitation. P03-P10 are outside that range and use zero recommended excitation for numerical safety in this current model.

The high-EA zero-input rows are not a full dynamic validation of those plants. They show the current dashboard safety policy. To dynamically validate P03-P10, the model should be supplied with the exact per-plant R, J, f, L, and b arrays plus retuned excitation limits.

8. Stored Artifacts

Type	Description	Path
PDF	Simulation comparison report	C:\Users\user\Documents\Git+ Learning\r2r-dashboard\reports\validation_summary\simulation_plant_excitation_report.pdf
CSV	50-row plant x excitation simulation sweep	C:\Users\user\Documents\Git+ Learning\r2r-dashboard\data\processed\simulation_plant_excitation_sweep.csv
JSON	Machine-readable report summary	C:\Users\user\Documents\Git+ Learning\r2r-dashboard\reports\validation_summary\simulation_plant_excitation_report.json
PNG	Tension RMSE heatmap	C:\Users\user\Documents\Git+ Learning\r2r-dashboard\reports\figures\simulation_tension_rmse_heatmap.png
PNG	Best-by-plant chart	C:\Users\user\Documents\Git+ Learning\r2r-dashboard\reports\figures\simulation_best_excitation_by_plant.png
PNG	Average-by-excitation chart	C:\Users\user\Documents\Git+ Learning\r2r-dashboard\reports\figures\simulation_average_by_excitation.png
PNG	EA scatter chart	C:\Users\user\Documents\Git+ Learning\r2r-dashboard\reports\figures\simulation_ea_vs_best_rmse.png